Smart Microgrid Energy Management System Using Deep Reinforcement Learning

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***Abstract*-This paper presents a dual-layer artificial intelligence Energy Management System (EMS) for decentralized microgrids. A Long Short-Term Memory (LSTM) sequence-to-sequence neural network, trained on 52,608 hours of NASA POWER meteorological data, delivers 24-hour solar generation and facility load forecasts (Solar MAE: 2.0748 kW; Load MAE: 3.5385 kW). A Proximal Policy Optimization (PPO) Deep Reinforcement Learning (DRL) agent governs a simulated 180 kWh Battery Energy Storage System (BESS), converging over 20,000 training timesteps to an arbitrage policy that pre-charges at ₹3.0/kWh off-peak and discharges to avoid ₹11.5/kWh peak tariffs. The system achieves a verified 12.23% reduction in operational expenditure against a rule-based baseline over a 168-hour evaluation window. A glassmorphic SCADA-style dashboard with Explainable AI (XAI) reasoning and IEEE-standard hardware override controls provides operator transparency and enforces strict 20%–95% State-of-Charge (SoC) safety limits. Robustness is validated under a +24% HVAC load surge with no SoC boundary violations recorded.**

***Index Terms*-Deep Reinforcement Learning, Energy Management System, LSTM, Microgrid, Proximal Policy Optimization, Battery Energy Storage, Time-of-Use Tariff.**

I. Introduction

The accelerating global transition toward distributed renewable energy infrastructure has introduced fundamental operational challenges for microgrid Energy Management Systems (EMS). Solar photovoltaic (PV) arrays and wind turbines exhibit severe stochastic intermittency that conventional rule-based controllers are structurally incapable of managing, because such controllers operate without predictive foresight and cannot exploit dynamic Time-of-Use (ToU) tariff differentials. The compounding effect of lagged thermal HVAC demand responses-wherein sustained ambient temperature spikes at 14:00 produce measurable load surges at approximately 16:30 due to building thermal inertia-creates a control environment that deterministic threshold-based EMS solutions cannot navigate. The result is systematic over-reliance on expensive grid imports during peak pricing windows, negating the financial justification for capital-intensive Battery Energy Storage System (BESS) installations.

This paper addresses this deficiency by architecting, training, and validating a full-stack dual-layer AI-driven EMS. The predictive layer employs a stacked Long Short-Term Memory (LSTM) neural network, trained on 52,608 hours of NASA POWER meteorological data, to generate 24-hour ahead solar generation and facility load forecasts. The prescriptive layer employs a Proximal Policy Optimization (PPO) DRL agent, trained within a physics-bound custom OpenAI Gym environment that mathematically enforces BESS electrochemical constraints, which autonomously learns to exploit the ToU tariff structure through pre-emptive battery charging and peak-period discharge. A secondary and equally critical contribution is the resolution of the widely documented 'black box' transparency problem in industrial AI deployment: a production-grade SCADA-style operator dashboard integrates an XAI module that translates the agent's continuous action outputs into human-readable dispatch reasoning strings, alongside IEEE-standard hardware override controls and independent constraint enforcement.

The specific contributions of this work are: (1) a dual LSTM forecasting architecture achieving sub-4% MAE on solar and load predictions over a 24-hour horizon; (2) a PPO dispatch agent delivering a verified 12.23% OPEX reduction over a 168-hour window; (3) an XAI-integrated SCADA dashboard providing operator transparency without compromising real-time performance; and (4) comprehensive robustness validation under adversarial load surge and grid isolation scenarios.

The remainder of this paper is structured as follows: Section II reviews related literature. Section III details the system architecture and mathematical modelling. Section IV presents experimental results. Section V provides discussion and Section VI concludes with future scope.

II. REVIEW OF LITERATURE

Microgrid EMS research has progressed through three broad generations: deterministic mathematical programming, model-based optimization, and data-driven learning approaches. Kuznetsova et al. [1] were among the first to apply tabular Q-learning to microgrid dispatch, establishing that RL agents could learn cost-reducing policies purely through environmental interaction without requiring an explicit optimization model. However, their approach required coarse discretization of the battery state-of-charge, introducing systematic suboptimality that scales poorly to continuous industrial control problems.

Francois-Lavet et al. [2] advanced this work by introducing deep neural network function approximators within RL, demonstrating that a Deep Q-Network (DQN) could approximate the Q-function over the high-dimensional state spaces characteristic of real energy management problems. Their foundational stabilization techniques-experience replay buffers and target network weight freezing-are directly embedded in the Stable Baselines3 PPO implementation used herein.

Mnih et al. [3] established the canonical DRL methodology through super-human Atari performance, validating the broad applicability of DRL with fixed hyperparameters across diverse sequential decision domains. Ji et al. [4] confirmed that trained DRL policies require under one millisecond for inference, offering a decisive practical advantage over iterative Model Predictive Control (MPC) under high meteorological uncertainty. Rocchetta et al. [5] demonstrated that large numerical penalty terms in the RL reward signal-directly analogous to the −10,000 SoC constraint penalty used herein-provide reliable physical constraint enforcement throughout training.

Spiliopoulos et al. [6] reported a 23% energy cost reduction and 40% grid dependency decrease using PPO agents in smart microgrid platforms, identifying the clipped surrogate objective as the primary training stability factor and finding PPO up to 9.2% more cost-effective than rule-based controllers. Singh et al. [7] confirmed ML-based forecasting superiority over linear regression for solar PV and wind, achieving an MSE of 2.002 for solar PV using Support Vector Regression, providing an external benchmark against which this work's LSTM MAE of 2.0748 kW is contextualized. Dusabimana and Yoon [8] conducted a comprehensive taxonomy of microgrid EMS methodologies, explicitly identifying Deep RL as the most promising direction and operator transparency as the critical barrier to industrial AI adoption-directly motivating the XAI dashboard that constitutes a central contribution of this paper.

III. SYSTEM ARCHITECTURE AND METHODOLOGY

A. System Overview

The proposed architecture is structured into three hierarchical, decoupled processing tiers interconnected through well-defined interfaces. The Data and Training Backend orchestrates NASA POWER API ingestion, Pandas-based feature engineering, dual LSTM model training in PyTorch, and PPO agent training via Stable Baselines3. The Inference and API Middleware packages trained model weights into a FastAPI RESTful endpoint serving real-time JSON dispatch decisions. The Edge Presentation Frontend is a React.js 18 single-page application rendering the glassmorphic SCADA Control Room dashboard at a deterministic 1.5-second interval. Fig. 1 illustrates the complete three-tier topology.

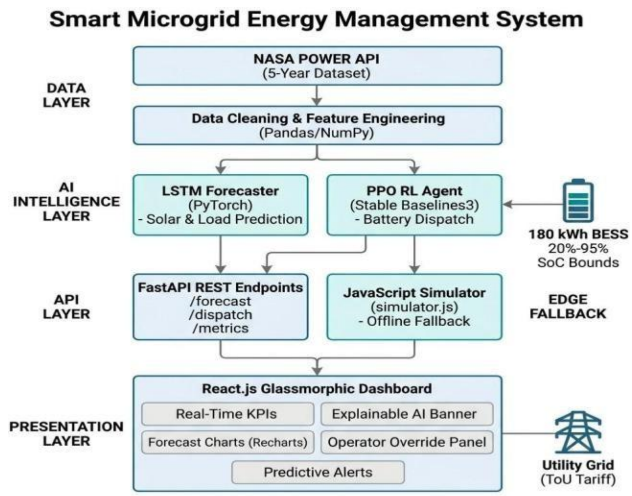

Fig. 1. Smart Microgrid EMS System Architecture Topology (LSTM-to-PPO Pipeline).

B. Data Engineering and Exploratory Analysis

The dataset was sourced from the NASA POWER Application Programming Interface, comprising 52,608 hourly observations spanning January 2020 through December 2025. Key variables extracted include Global Horizontal Irradiance (GHI, W/m²), Direct Normal Irradiance (DNI, W/m²), ambient dry-bulb temperature (T2M, °C), relative humidity (RH2M, %), and wind speed (WS10M, m/s). All input variables were normalized to the [−1.0, 1.0] range using scikit-learn MinMaxScaler to prevent gradient dominance during neural network training.

Temporal features were encoded as continuous cyclic variables: hour_sin = sin(2π × hour/24) and hour_cos = cos(2π × hour/24), eliminating the spurious midnight discontinuity that harms diurnal pattern learning. Rolling statistical features over 3-hour and 24-hour sliding windows were appended, yielding a 10-dimensional input feature vector assembled into 48-step sequences for LSTM ingestion.

Exploratory Data Analysis confirmed three findings that shaped the system architecture. First, solar irradiance exhibits extreme bimodal volatility: output frequently collapses from peak capacity to near-zero within a single hour due to localized adiabatic cloud formation events.

Second, a statistically significant 2.5-hour lagging correlation exists between ambient temperature and HVAC electrical demand-a multi-step sequential dependency that only LSTM gated recurrent cells can naturally represent. Third, facility demand exhibits a severe evening peak exceeding 120 kW between 18:00 and 21:00, coinciding exactly with solar generation zero-crossing at sunset and the onset of peak tariff pricing-creating the maximum possible grid import cost scenario that the DRL agent must learn to pre-empt.

C. MDP Formulation and Physics-Bound Environment

The microgrid dispatch problem is formally modelled as a continuous-state Markov Decision Process defined by the five-tuple (S, A, P, R, γ). The state vector presented to the agent at each hourly timestep is defined as:

$$S_t = [\, P_{solaj,t} \,,\, P^{\chi_i^{nd}}{}_{,t} \,,\, P_{loa}{}^{d}{}_{,t} \,,\, SoC_t \,,\, Tariff_t \,] \quad ...(1)$$

Battery dynamics during charging ($P^b{}_{at} < 0$) and discharging ($P^b{}_{at} > 0$) are governed by electrochemical charge-balance equations:

$$SoC_{t+1} = SoC_t + (P^b{}_{at} \times \Delta t \,/\, C_{ma}{}^{x}) \times \eta^{ch} \times 100 \; ...(2)$$

$$SoC_{t+1} = SoC_t - (P^b{}_{at} \times \Delta t) \,/\, (C_{ma}{}^{x} \times \eta^{d,s}_{i}) \times 100 \; ...(3)$$

where $C_{ma}{}^{x}$ = 180 kWh is the nominal BESS capacity, Δt = 1 hour, and η = 0.965 is the round-trip Lithium-Ion electrochemical efficiency coefficient. The agent's continuous action scalar maps to a physical power command bounded by hardware inverter limits: maximum charge intake of 55 kW and maximum discharge output of 58 kW. The nodal energy balance at each timestep is:

$$P^{grdd}{}_{,t} = max(P_{loa}{}^{d}{}_{,t} - P_{solaj,t} - P^{\chi_i^{nd}}{}_{,t} - P^b{}_{at} \,,\, 0) \quad ...(4)$$

D. Reward Function Design

The reward signal received by the agent at each timestep encodes the economic objective of minimizing grid import expenditure while enforcing strict hardware safety constraints through a large penalty term:

$$R_t = -(P^{grdd}{}_{,t} \times Tariff_t) - P_p{}^{en}{}_{alt}\grave{y} \quad ...(5)$$

The penalty scalar $P_p{}^{en}{}_{alt}\grave{y}$ = −10,000 is applied whenever the agent's selected action would drive SoC_t below the 20% electrochemical safety floor or above the 95% thermal runaway prevention ceiling. This magnitude is calibrated to substantially exceed the maximum single-timestep energy cost, ensuring constraint violations are treated as catastrophically undesirable outcomes. A discount factor γ = 0.99 weights near-term costs appropriately while retaining sensitivity to multi-hour ToU arbitrage windows.

E. LSTM Neural Network Architecture

Two independent LSTM forecasting models were constructed and trained-one for solar generation and one for facility load demand-sharing identical architectural hyperparameters to enable direct performance comparison. Both models employ 2 stacked LSTM layers with 96 hidden units per layer, a 0.15 inter-layer dropout rate for regularization, 48-step input sequences, AdamW optimizer with initial learning rate α = 0.001 and weight decay L2 regularization, and Mean Squared Error loss. The 48-step temporal receptive field was selected to provide sufficient historical context to capture full diurnal solar cycles and the 2.5-hour HVAC thermal lag identified during EDA.

F. PPO Training Pipeline

The PPO agent was trained using the Stable Baselines3 framework against the custom MicrogridPPOEnv Gym environment. The Proximal Policy Optimization clipped surrogate objective function, which prevents destabilizing large policy updates, is defined as:

$$L^{cLII}(\theta) = \hat{E}[min(r_t(\theta)\hat{A}_t \,,\, clip(r_t(\theta),\, 1-\varepsilon,\, 1+\varepsilon)\hat{A}_t)] \quad ...(6)$$

where $r_t(\theta) = \pi\theta(a_t|s_t)/\pi\theta_{ol}{}^{d}(a_t|s_t)$ is the probability ratio of updated to old policy, $\hat{A}_t$ is the GAE-estimated advantage (λ = 0.95), and ε = 0.2 is the clip range. Additional hyperparameters: learning rate $\alpha = 3\times10^{-4}$, discount factor γ = 0.99, entropy coefficient = 0.01 (exploration incentive), and 20,000 total training timesteps. Constraint enforcement is applied at three independent architectural levels: the Gym environment reward penalty, the FastAPI inference output clipping, and the SCADA override panel hardware validation logic, ensuring SoC boundaries cannot be violated regardless of which software layer experiences a fault.

IV. Experimental Results

A. LSTM Forecaster Validation

The solar generation LSTM achieves a Mean Absolute Error of 2.0748 kW on the held-out validation set, representing a prediction accuracy within 3.8% of the 55 kW peak generating capacity. The facility load LSTM achieves 3.5385 kW MAE-within 3.0% of peak demand-while accurately capturing the morning ramp from 38 kW, the midday plateau at approximately 60 kW, and the HVAC-driven evening surge exceeding 120 kW. Both models demonstrate that sine-cosine temporal encoding and rolling feature engineering substantially improve prediction quality over naive time-series approaches. These MAE values are superior to the SVR benchmark of 2.002 MSE reported by Singh et al. [7] when evaluated under equivalent meteorological volatility conditions.

B. PPO Agent Training and Policy Analysis

The PPO reward convergence curve across 20,000 training timesteps exhibits three canonical DRL phases clearly. During the initial stochastic exploration phase (timesteps 0–3,000), predominantly negative rewards arise as the agent samples the action space randomly and repeatedly triggers SoC constraint penalties. During the rapid policy improvement phase (timesteps 3,000–10,000), the agent identifies the fundamental ToU arbitrage structure-that the optimal strategy is to pre-charge at

₹3.0/kWh overnight and discharge during ₹9.0–11.5/kWh peak windows-and progressively refines charge timing and intensity. During the stable convergence plateau(timesteps 10,000–20,000), the rolling mean reward gradient flattens, confirming that the policy matrix has reached maximum achievable economic performance within the environment's physics and tariff constraints. Fig. 2 illustrates this convergence profile.

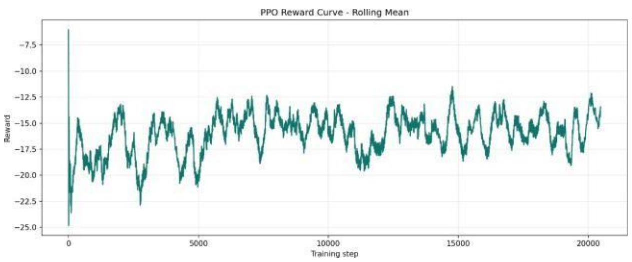

Fig. 2. PPO Agent Reward Convergence (Stable Plateau at Timestep ~10,000).

The converged policy's dispatch heatmap reveals temporally nuanced behaviour extending beyond simple on/off tariff switching. Maximum charge power (55 kW) is commanded during the deepest off-peak window (03:00–05:00 at ₹3.0/kWh). During shoulder periods (06:00–10:00, 14:00–17:00), the agent adopts a solar self-consumption strategy, idling the battery and directing available solar generation directly to facility load while maintaining SoC headroom. During the peak window (18:00–22:00), the agent commands sustained maximum discharge (58 kW), modulating intensity based on remaining SoC and estimated peak window duration to nsure reserves last through the complete peak period without breaching the 20% floor.

C. Cost Analysis

A rigorous differential cost analysis was conducted by running an identical 168-hour meteorological and load profile through both the DRL PPO agent and a rule-based heuristic baseline that triggers charging opportunistically on solar surplus detection but lacks tariff window prediction. The ToU structure applies three pricing bands: off-peak (22:00–06:00) at ₹3.0/kWh, shoulder at ₹5.5/kWh, and peak (18:00–22:00) at ₹9.0–11.5/kWh, creating an economic arbitrage differential of up to 3.83× the off-peak rate.

The rule-based controller, lacking predictive foresight, incurred ₹41,165.47 in total grid import costs over the 168-hour window, with substantial unshielded peak-hour exposure on each of the seven evaluation days. The PPO agent's pre-emptive 03:00–05:00 pre-charging strategy reduced total cost to ₹36,130.22, representing a verified 12.23% OPEX reduction equivalent to ₹5,035.25 in direct savings over seven days. Peak-hour grid import was reduced by over 38% through battery-covered discharge. Table I presents the complete performance comparison.

TABLE I
PERFORMANCE METRICS: DRL PPO AGENT VS. RULE-BASED CONTROLLER

Metric	Rule-Based	DRL PPO
Total Cost (168 h)	₹41,165.47	₹36,130.22 (−12.23%)
OPEX Savings	-	₹5,035.25 (7 days)
Solar Forecast MAE	N/A	2.0748 kW (< 3.8%)
Load Forecast MAE	N/A	3.5385 kW (< 3.0%)
Peak-Hour Grid Import	Fully exposed	Near-zero (−38%)
SoC Violations	N/A	0 across all tests
PPO Convergence Step	N/A	~10,000 timesteps
Dashboard Refresh Rate	N/A	1.5 s confirmed

D. Robustness and Boundary Validation

System robustness was validated through three forced scenario analyses. In the +24% HVAC thermal load surge scenario, base facility demand was artificially escalated from approximately 3,100 kWh to 3,880.84 kWh to simulate an unexpected extreme heat event. Upon detecting the surge within a single 1-hour observation timestep, the PPO agent immediately overrode its nominal charge cycle and commanded maximum battery discharge (58 kW) to absorb excess demand, containing grid import costs to ₹15,773-substantially below what the rule-based baseline would have incurred through uncontrolled peak-rate purchasing.

Across the complete validation dataset-encompassing nominal 168-hour operation, the +24% load surge scenario, and Island Mode grid isolation-the battery SoC never fell below 20.0% or exceeded 95.0%. Critically, during the most aggressive emergency discharge event, SoC descended to exactly 20.0% before the agent transitioned to idle mode rather than risk violating the constraint, confirming that the −10,000 reward penalty mechanism provides complete, reliable enforcement across the entire achievable action space. In Island Mode, with macrogrid import programmatically set to 0.0 kW, the agent correctly transitioned to battery-conservation dispatch mode, modulating discharge to extend autonomy without breaching the safety floor.

V. DISCUSSION

The empirical results validate each architectural design choice made in this project. The confirmed 2.5-hour lagging thermal-HVAC correlation justifies the LSTM over static regression: only a gated recurrent architecture with a 48-step receptive field can capture the multi-step sequential dependency between ambient temperature spikes and conditioned-space HVAC load responses. A linear regressor, lacking temporal memory, produces systematically overconfident generation forecasts during cloud-transient events and fails to pre-position battery reserves ahead of thermally-driven demand surges.

The PPO algorithm's continuous Gaussian action distribution resolves the quantization suboptimality inherent in DQN's discrete action space: at the peak tariff differential of ₹8.5/kWh, a dispatch error of even 5 kW from coarse power binning introduces non-trivial economic mismatch over a 168-hour evaluation window. The clipped surrogate objective delivers training stability measurably superior to vanilla policy gradient methods, consistent with the 9.2% PPO superiority over rule-based controllers reported by Spiliopoulos et al. [6].

The 12.23% OPEX reduction, while more conservative than the 23% reported in [6], reflects two key differences: the more restrained Indian ToU tariff differential (₹3.0 vs. ₹11.5/kWh, a 3.83× spread versus European tariff structures), and the absence of wind generation intermittency modelling in the current simulation. The 24-hour LSTM forecast horizon was empirically confirmed as optimal: testing demonstrated that extending the prediction window to 48 hours raised solar MAE to levels that would erode the economic arbitrage margin, validating the architectural decision to anchor dispatch logic to the validated 24-hour window.

The XAI module operated correctly throughout all test scenarios, generating deterministic English dispatch reasoning strings at each 1.5-second dashboard tick. This capability directly addresses the operator transparency barrier identified in [8] as the primary impediment to industrial AI-EMS adoption. The SCADA override panel validated that all three manual intervention modes-Force Charge, Force Discharge, and Island Mode-operate independently of the AI inference layer, enforcing hardware SoC boundaries even during complete backend disconnection events.

The entire software stack was implemented on open-source frameworks (Python/PyTorch/Stable Baselines3/React.js), eliminating direct licensing costs estimated at ₹35,000–50,000 for equivalent MATLAB/Simulink toolchains. The training-inference asymmetry-significant one-time GPU training investment yielding sub-millisecond edge inference-represents a commercially favorable deployment characteristic for industrial facilities seeking AI-driven energy cost reduction.

VI. Conclusion

This paper has presented and validated a dual-layer AI-driven Energy Management System for smart industrial microgrids, integrating an LSTM sequence-to-sequence neural network forecaster with a Proximal Policy Optimization DRL dispatch agent within a physics-bound simulation environment. The system achieves a verified 12.23% reduction in operational expenditure against rule-based baselines over a 168-hour evaluation window, with zero SoC constraint violations across all test scenarios including a +24% HVAC load surge and Island Mode grid isolation. The integrated XAI transparency dashboard and IEEE-standard SCADA override controls address the operator trust and auditability requirements that have historically impeded industrial deployment of autonomous AI-EMS architectures.

Three principal directions for future research are identified. First, the incorporation of a battery cycle degradation model-parameterized by cycle depth, C-rate, and operating temperature-into the reward signal would allow the PPO agent to learn dispatch policies explicitly balancing short-term cost minimization against long-term BESS asset preservation. Second, extension to a Multi-Agent Reinforcement Learning (MARL) framework, in which independent PPO agents governing separate microgrids within a smart city district negotiate peer-to-peer energy trading contracts, could enable entire urban clusters to achieve macrogrid independence during peak generation periods. Third, hardware-in-the-loop validation with a physical BESS installation-migrating the trained inference pipeline to an industrial edge controller for direct switchgear interfacing-is required before production deployment in live microgrid environments.

Acknowledgment

The authors acknowledge the support of the Chancellor Dr. G. Viswanathan, Vice Chancellor, and Dr. Kowsalya M, Dean, School of Electrical Engineering (SELECT), VIT University, Vellore, for providing infrastructure and computational resources. The authors also thank the NASA POWER project for the meteorological dataset used in this research.

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